Zewail City of Science and Technology
University of Science and Technology
School of Computational Sciences and Artificial Intelligence
CSAI 202 - Fall 2026

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Introduction to Software Engineering
Academic Course & Schedule Planer
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1. Introduction

1.1 Purpose

This SRS document details the functional and non-functional specifications for our Academic Course and Schedule Planner web application. The primary objective of our document is establishing an agreement between the project stakeholders, academic admins, and the development team in the system's capabilities and boundaries before designing, and coding. Serving as the baseline for our system design, MVC structuring, and unit testing.

1.2 Scope

Our Academic Course & Schedule Planner is a centralized web platform designed to streamline semester registration planning and academic management for university students and admins.

Our system provides tools for students to build conflict-free weekly schedules, track the prerequisite trees, estimate future GPAs, section schedule generation, and receive notifications. And for the administrative staff, the platform offers control over the courses, capacity in sections, and prerequisite rules.

Core Goals & System Capabilities:

Eliminating scheduling overlaps through automated timetable generation engines.
Preventing invalid registrations in real-time

1.3 Definitions, Acronyms, and Abbreviations

SRS: Software Requirements Specification

GPA: Grade Point Average

MVC: Model-View-Controller (Software Architectural Pattern)

UML: Unified Modeling Language

UI/UX: User Interface / User Experience

API: Application Programming Interface

HTTP/HTTPS: Hypertext Transfer Protocol / Hypertext Transfer Protocol Secure

1.4 References

IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications.

Course Syllabus & Project Specification for CSAI 203: Introduction to Software Engineering, Zewail City of Science and Technology, Fall 2026.

IEEE Standard 1063-2001: IEEE Standard for Software User Documentation.

1.5 Overview

The remainder of this SRS document is structured as follows:

Section 2 (Overall Description): Provides context regarding our product perspective, user classifications, operational environment, design constraints (MVC requirement), user manuals, and system assumptions.

Section 3 (Specific Requirements): Details 20 functional requirements, UML Use Case diagrams with IEEE templates, a conceptual domain class diagram, testable non-functional requirements with success criteria, and interface constraints.

Section 4 (Appendices): Contains the system Data Dictionary and technical Glossary

2. Overall Description

2.1 Product Perspective

Our Academic Course & Schedule Planner is part of "University Student Portal", designed to support students and administrators in academic planning and course scheduling. The system works alongside other university services such as student information management, course registration and academic records management. The application allows students to plan their semesters, create conflict-free timetables, check prerequisite requirements and calculate their GPAs. The university administrators can manage course offerings, section capacities, schedules and prerequisite information.

Our system exchanges data with other university subsystems, including student records, course catalogs and academic results to provide accurate and consistent information. However, the actual integration depends on the availability of these subsystems and their interfaces.

Our Academic Course & Schedule Planner focuses specifically on academic planning and scheduling. It does not replace the entire University Management System instead,

it provides specialized features that complement the other components of the larger system.

2.2 Product Function

1. Login
2. Search
3. Register/Remove
4. Prerequisite
5. Conflict-free
6. Schedule (Viewing weekly timetable)
7. Calculate semester GPA
8. Calculate cumulative GPA (cGPA)
9. Course capacity
10. Multiple schedule generator
11. Calculate credit hour
12. Section schedule
13. Managing courses by admin
14. Elective courses (Info)
15. Student transcript
16. Contact academic advisor
17. Instructors' office hours
18. View course details
19. Attendance tracking
20. Grade update notifications

2.3 User Classes and Characteristics

1. Student:
    - Register courses
    - Check prerequisites
    - Access the transcript
2. Admin:
    - Manage the course
    - Update course capacity
    - Grade update

2.4 Software Stakeholders

1. Student: right schedule, privacy of grades
2. Admins: Consistent offerings, controlled capacities, fewer manual corrections.

3. Academic Advisor: Students following program plans and demand information per course.
4. Developing team & testing team: Frontend, backend and database implementations.

2.5 Operating Environment

1. Hardware: Laptops, computers and mobiles
2. OS: Windows, Mac and android
3. Browsers: All web browsers
4. Network: TCP/IP over HTTP

2.6 Design and Implementation Constraints

Our Academic Course & Schedule Planner will be developed as a web based application using programming languages and tools selected by project team. Our system will use a database to store and manage academic information. It must enforce role based access control to protect the students' and the admins' data. The system must validate prerequisites, prevent timetable conflicts and calculate GPAs according to the defined academic rules. The design must also follow the project requirements and the formatting guidelines specified by the course instructor.

2.7 User Documentation

Our system will provide a (FAQ) section to help students and administrators understand how to use the application. It will include guidance on course selection, timetable creation, prerequisite checking, GPA calculation and course management.

2.8 Assumptions and Dependencies

Our system assumes that users have internet access and a compatible web browser. It also assumes that course information, schedules, capacities, prerequisites and academic records are accurate and up to date. The system depends on a web server, a database management system and the availability of these services. Integration with other university systems is dependent on the availability of their data and interfaces

3. Specific Requirements

3.1 Functional Requirements

FR1: Login

Our system allows students and admins to log in using their registered credentials. Our system verifies the entered information and provides access according to the user's role.

FR2: Search Courses

Our system allows students to search for courses by course name, course code, department or instructor. The system displays matching courses and their basic information.

FR3: Register and Remove Courses

Our system allows students to register for available course sections or remove registered courses during the permitted registration period. The system updates the student's registered courses after each successful operation.

FR4: Prerequisite Checking

Our system checks whether students have completed the required prerequisite courses before allowing them to register for a course. If the prerequisites are not satisfied, the system informs the student.

FR5: Conflict-Free Timetable

Our system verifies that a student's selected course sections do not have overlapping lecture times. If a conflict is detected, the system notifies the student and prevents the conflicting sections from being added to the same timetable.

FR6: Weekly Timetable Viewing

Our system generates and displays a weekly timetable showing the students' registered courses, lecture days, times, and locations. Students are able to view their schedules in an organized format.

FR7: Calculate Semester GPA

Our system calculates the student's semester GPA based on the grades and credit hours of courses taken during the selected semester, according to the university's grading rules.

FR8: Calculate Cumulative GPA (cGPA)

Our system calculates the student's cGPA using the academic results and credit hours of all completed semesters, according to the university's grading rules.

FR9: Course Capacity

Our system displays the maximum capacity and available seats for each course section. The system prevents students from registering for sections that have reached their capacity.

FR10: Multiple Schedule Generator

Our system generates multiple possible timetables based on the students' selected courses and available sections. Each generated timetable avoids timetable conflicts and respect course prerequisites and section capacities.

FR11: Calculate Credit Hours

Our system calculates and displays the total credit hours of the courses selected by a student. The system also notifies the student if the selected credit hours exceed the university's minimum or maximum allowed limits.

FR12: Section Schedule

Our system displays the schedule of each course section, including lecture days, start and end times, locations, and assigned instructors. Students are able to compare sections before registering.

FR13: Course Management by Administrator

Our system allows administrators to add, update, and remove course information, create course sections, assign instructors, set capacities, and define prerequisites. Our system ensures that changes are reflected in the information available to students.

FR14: Elective Course Information

Our system provides information about elective courses, including course descriptions, credit hours, departments, and any applicable requirements. Students are able to review this information before selecting electives.

FR15: Student Transcript

Our system allows students to view their academic transcripts, including completed courses, grades, credit hours, and semester results. Access to transcript information is restricted to authorized users.

FR16: Contact Academic Advisor

Our system provides students with the contact information of their assigned academic advisors. It allows students to access the available communication details to ask for academic guidance.

FR17: Instructor Office Hours

Our system displays the instructors' office hours, times, and office locations. Students are able to use this information to plan meetings with their instructors.

FR18: View Course Details

Our system allows students to view detailed information about each course including code, name, description, credit hours, prerequisites and available sections.

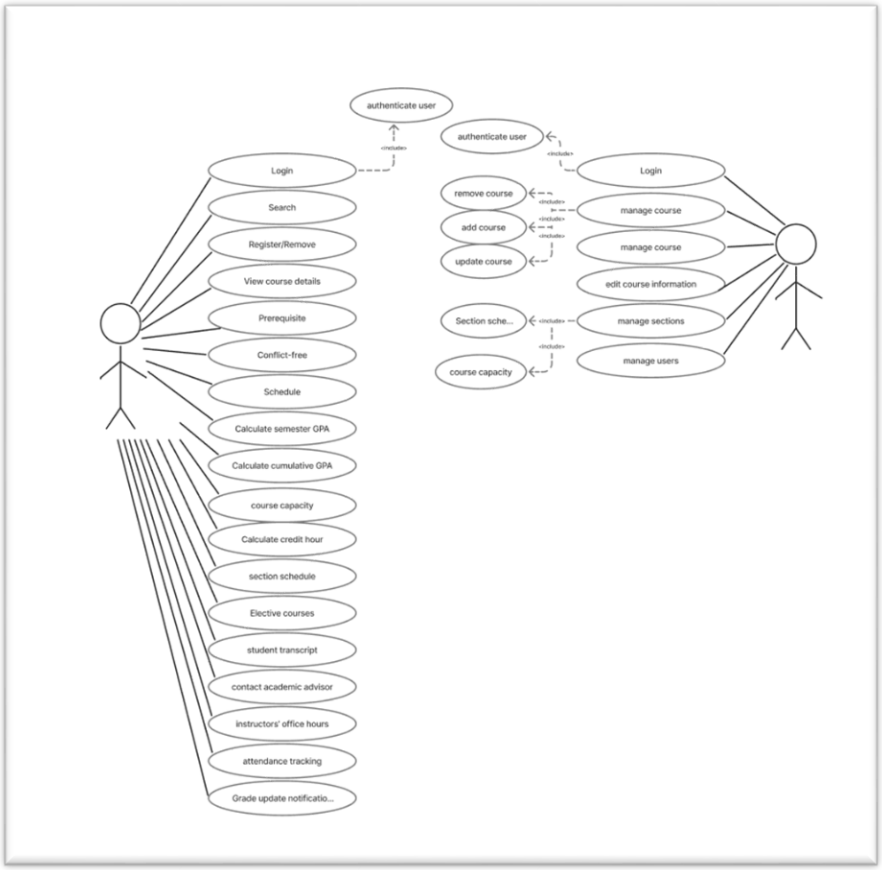
FR19: Attendance Tracking

Our system allows authorized instructors or administrators to record and update student attendance for course sessions. Students are able to view their attendance records for their registered courses.

FR20: Grade Update Notifications

Our system notifies students when their grades are added or updated. The notification identifies the relevant course and indicate that the student's academic record has changed.

3.2 Use CaseModel



3.2.1 Use Case Diagram

3.2.2 Use Case Description

ID	Name	Description	Actor	Preconditions	Postconditions	Main flow
UC 1	Login	Allows students and admins to access the system using their credentials.	Student - Admin	The actor has an existing account.	The actor is logged in and can access authorized functions.	1. The actor opens the login page. 2. Enters username/email and password. 3. The system validates the credentials
UC 2	Search	Allows students to search for courses using course	Student	The course catalog is available.	Relevant course search results are displayed.	1. The student opens the search page. 2. Enters search criteria. 3. The

		names, codes, or keywords.				system searches the course catalog.
UC 3	Register/ Remove	Allows students to register for available courses or remove courses from their registered list.	Student	The student is logged in and the registration period is open.	The student's registered courses are updated if the request succeeds.	1. The student views available courses. 2. Selects a course to register for or remove. 3. The system checks prerequisites, capacity, and timetable conflicts when registering.
UC 4	Prerequisite Checking	Allows students to view and verify the prerequisite requirements for a course.	Student	Course prerequisite information is available.	The student understands the prerequisite requirements and eligibility status.	1. The student selects a course. 2. The system retrieves its prerequisites. 3. The system compares the requirements with the student's academic record.
UC 5	Conflict-Free Schedule Checking	Checks whether selected course sections have overlapping class times.	Student	Course section days and times are available.	The student receives the schedule conflict-checking result.	1. The student selects courses or sections. 2. The system retrieves their meeting times. 3. The system compares the time intervals.
UC 6	Schedule	Allows students to view their registered courses in a	Student	The student is logged in.	The student can view their weekly academic schedule.	1. The student opens the schedule page. 2. The system retrieves registered

		weekly timetable.				course sections. 3. The system organizes classes by day and time
UC 7	Calculate Semester GPA	Calculates the student's Grade Point Average for a selected semester.	Student	Semester course grades and credit hours are available.	The semester GPA is displayed when sufficient data is available.	1. The student selects a semester. 2. The system retrieves course grades and credit hours. 3. The system calculates the weighted grade points.
UC 8	Calculate Cumulative GPA	Calculates the student's cumulative Grade Point Average across completed semesters.	Student	Academic records from the relevant semesters are available.	Academic records from the relevant semesters are available.	1. The student selects the cumulative GPA function. 2. The system retrieves eligible course grades and credit hours. 3. The system applies the institution's GPA
UC 9	Generate Multiple Schedules	Generates multiple possible timetables based on the student's selected courses and scheduling preferences.	Student	Course sections and their schedules are available.	One or more valid schedule options are available for the student to review.	1. The student selects a course section. 2. The system retrieves the maximum capacity and current enrollment. 3. The system calculates the number of available seats. 4. The system displays the result.

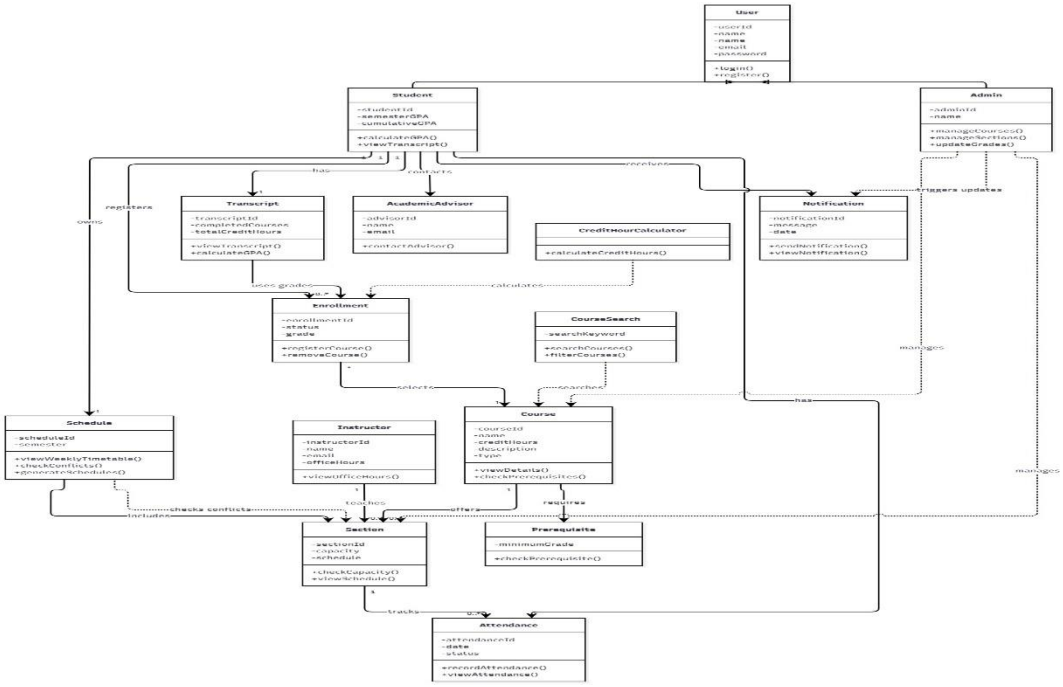
UC 10	Calculate Credit Hours	Calculates the total credit hours associated with the student's selected or registered courses.	Student	Credit-hour values are defined for the selected courses.	The total credit hours are displayed	1. The student opens the credit-hour calculator. 2. The system retrieves the selected courses. 3. The system adds their credit hours
UC 11	Generate Multiple Schedules	Generates multiple possible timetables based on the student's selected courses and scheduling preferences.	Student	Course sections and their schedules are available.	One or more valid schedule options are available for the student to review.	1. The student selects the desired courses. 2. The student enters scheduling preferences, if supported. 3. The system retrieves available sections. 4. The system generates possible schedule combinations
UC 12	View Section Schedule	Allows students to view the schedule details of individual course sections.	Student	Section schedule information is available.	The student can review the selected section's schedule.	1. The student selects a course. 2. The system displays its available sections. 3. The student selects a section. 4. The system displays the section's days, times, location, and instructor, where available.

UC 13	Manage Courses	Allows the admin to create, update, and remove course information in the system.	Admin	The admin is logged in and authorized to manage courses.	The course catalog reflects the successful changes.	1. The admin opens course management. 2. Selects an operation: add, update, or remove. 3. Enters the required information or selects an existing course.

3.3 Domain Model

3.3.1 Conceptual Class Diagram

3.3.2



Class Descriptions

Class	Attributes	Relationship	Purpose
User	userId, name, email and password	Parent class of Student and Admin	Stores common user information and login details.
Student	studentId, semesterGPA and cumulativeGPA	Registers for Enrollment, owns Schedule and Transcript, contacts	Manage student registration and academic information.

		AcademicAdvisor, receives Notification and has Attendance records.	
Admin	AdminId and name	Manage Course and Section and trigger Notification.	Manages courses, sections and grade updates.
Course	courseId, name, creditHours and description	Has Sections, require Prerequisite and search through CourseSearch.	Stores course details and credit hour information.
CourseSearch	searchKeyword	Searches and filters Course.	Allows students to search for courses and view course details.
prerequisite	minimumGrade	Associated with Course	Defines requirements that must be completed before course registration.
Section	sectionId, capacity and schedule	Belongs to Course, taught by Instructor, and included in Schedule and Attendance records.	Represents a course section, capacity and timetable.
Enrollment	enrollmentId, status and grade	Connects Student with Course and provides information for Transcript and CreditHourCalculator.	Manages course registration, removal, and enrollment status.
Schedule	ScheduleId and semester	Owned by Student and include Section.	Displays weekly timetables, checks conflicts and supports schedule generation.
Transcript	transcriptId, completedCourses, totalCreditHours	Belongs to Student and uses Enrollment information.	Displays academic history, completed courses, grades and GPA.
CreditHourCalculator	None	Uses Enrollment information.	Calculates the total credit hours of registered courses.
Instructor	instructorId, name, email and officeHours	Teaches multiple Sections.	Stores instructor information and office hours.

Attendance	attendanceId, date and status	Associated with Student and Section.	Records and tracks student attendance.
AcademicAdvisor	advisorId, name and email	advisorId, name, email	Provides advisor information for academic guidance.
Notification	notificationId, message and date	Received by Student and triggered by Admin for grade updates.	Notifies students about important updates, including grade changes.

3.4 Non-Functional Requirements

Non-Functional Requirement	Description	Testing Method
Performance	The system should respond to searches and schedule requests in less than 3 seconds.	Check how long searches and schedule requests take.
Security	Users must log in and each user can only access their allowed features.	Test login and check that students cannot access admin features.
Usability	The system should be easy to use for students and admins.	Ask students to search for and register for a course.
Availability	The system should be available throughout the semester including the registration period.	Check that students can access the system during the semester and registration period.
Compatibility	The system should work on all browsers.	Open the system in each browser and test its features.

3.5 External Interface Requirements

3.5.1 User Interface

- Login: Secure access for students and admins.
- Course Management: Search, view, register, and remove courses.
- Schedule: View weekly timetables, check conflicts, and generate multiple schedules.
- Academic Calculations: Calculate semester GPA, cumulative GPA, and credit hours.

- Academic Information: View prerequisites, electives, course capacity, section schedules, and transcripts.
- Student Services: View attendance, contact academic advisors, and check instructors' office hours.
- Notifications: Display grade update notifications.
- Admin Dashboard: Allow admins to manage course information.

3.5.2 Hardware Interface

- Computers and Laptops: Access through a web browser.
- Mobile Devices: Access through smartphones and tablets.
- Input Devices: Use of keyboards, mice, and touchscreens.
- Network Devices: Internet connectivity to access the system.

3.5.3 Software Interface

- Flask Backend: The system will use Flask to handle user requests, course registration, schedule management and GPA calculations.
- Database: The system will store student information, courses, sections, grades and schedules

3.5.4 Communication Interface Network Protocol

Communication runs exclusively over the HTTP protocol over TCP port 443

4. Appendices

4.1 Appendix A: Data Dictionary

	Field	Constraints	Description
Users	user_id	Primary Key, Auto Increment	Unique user identity key
	email	Unique, Not Null	Login email address
	password_hash	Not Null	Bcrypt hashed account password
	role	'STUDENT', 'ADMIN'	Account permission tier
Courses	course_id	Primary Key	Course identifier code

	title	Not Null	Official course name
	credit_hours	Check (credit_hours > 0)	Assigned academic credit value
Sections	section_id	Primary Key, Auto Increment	Unique section offering key
	course_id	Foreign Key (Courses.course_id)	Associated parent course
	capacity	Check (capacity >= 0)	Maximum enrollment limit
	enrolled_count	Check (enrolled_count <= capacity)	Active enrolled student total
	day_time_slot	Not Null	Meeting schedule
Prerequisites	course_id	Foreign Key (Courses.course_id)	Target course requiring prerequisite
	prereq_id	Foreign Key (Courses.course_id)	Required prerequisite course code

4.2 Appendix B:

Glossary

Conflict-Free Timetable: A weekly class schedule containing no overlapping lecture, lab, or tutorial time slots. Prerequisite: A specific course or academic condition that must be successfully completed prior to enrolling in a higher-level course. Cumulative GPA: The overall grade point average computed by dividing total weighted grade points earned by total attempted credit hours. Section Capacity: The maximum number of students permitted to enroll in an individual class section. Model-View-Controller (MVC): An architectural pattern separating application logic into Models (data), Views (user interface), and Controllers (business logic execution).